Theory Week 14

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14	16/12/24 18/12/24	waves in rods, local LMB, eigenfrequencies and mode shapes Christmas lecture (final review with many experiments) *exercise 14: local balance of linear momentum, waves and vibrations in rods*	5.2-5.3.2

Orga

Mechanics 3 CAB G59

WhatsApp group

- Find all doc's on my website (link in Whatsapp bio)
- remember to do the STACK
- Study Center
  → Wednesday 18:00-20:00 (ETF C1)
- Summary videos from Dennis on moodle

5.2 Deformable bodies with non-negligible mass

Bodies no longer move as rigid bodies → Movement no longer easily described by motion of its center of mass (& angular vel. / acc.) Instead each point can undergo a different trajectory.

→ can formulate the kinetic relations for global (entire body) or macroscopic level

LMB

global balance of linear momentum for bodies and sub-bodies:

$$\sum_i \boldsymbol{F}_i^{\text{ext}} = M\boldsymbol{a}_{\text{CM}}$$

local balance of linear momentum:

stress components

$$\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{f} = \rho\boldsymbol{a} \qquad \text{or} \qquad \sum_{j=1}^{3}\frac{\mathrm{d}\sigma_{ij}}{\mathrm{d}x_j} + f_i = \rho a_i \qquad \text{for} \quad i = 1,2,3$$

body force density

5.3 Waves & Vibrations in slender rods

longitudinal wave equation for stretching/compression of a homogeneous slender bar:

$$\ddot{u}(x,t) = c^2 u_{,xx}(x,t) \qquad \text{with} \qquad c = \sqrt{\frac{E}{\rho}}$$

torsional wave equation for twisting of a homogeneous slender bar:

* $$\ddot{\theta}(x,t) = c_{\mathrm{T}}^2 \theta_{,xx}(x,t) \qquad \text{with} \qquad c_{\mathrm{T}} = \sqrt{\frac{G}{\rho}}$$

general solution for **longitudinal vibrations** (and **torsion** analogously):

$$u(x,t) = \hat{u}(x)q(t) \qquad \text{with}$$

$$\hat{u}(x) = B_1\cos\left(\frac{\omega}{c}x\right) + B_2\sin\left(\frac{\omega}{c}x\right) \qquad \text{and} \qquad q(t) = A_1\cos(\omega t) + A_2\sin(\omega t)$$

$\Delta l = u_x(l)$

$u(x,t)$ = displacement field

c = longitudinal wave speed

As there are many eigenfrequencies (ω) ⇒ complete soln is sum of all! (Analysis 3)

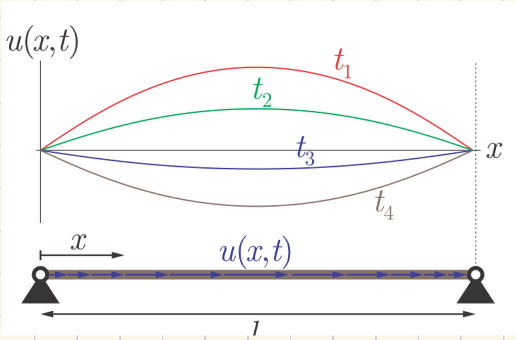
complete solution for **longitudinal vibrations** (and **torsion** analogously):

$$u(x,t) = \sum_{n=1}^{\infty}\hat{u}_n(x)q_n(t) \qquad (\text{torsion analogy: } u \leftrightarrow \theta,\ c \leftrightarrow c_{\mathrm{T}})$$

↳ The lowest non-zero eigenfrequency ω_1 = natural fundamental frequency

To find constants we need to know the configuration of the bar.

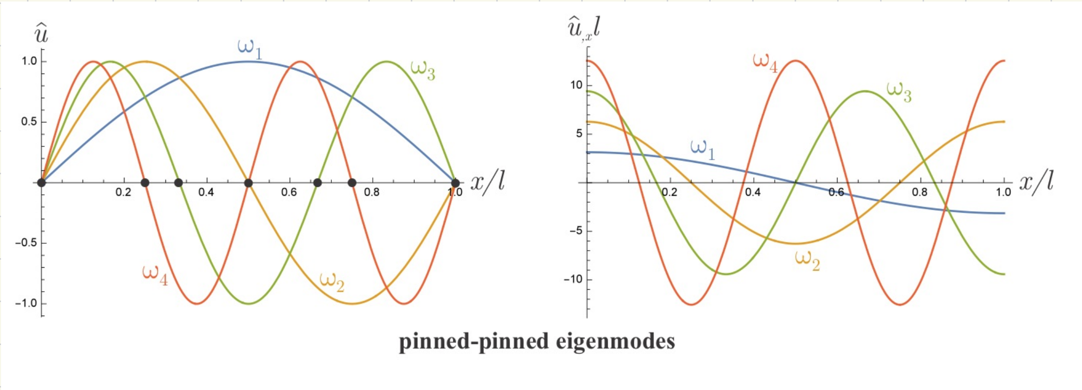
S.217



$$\Longrightarrow \quad u(0,t) = u(\ell,t) = 0 \quad \forall t$$
$$\hat{u}(0)q(t) = \hat{u}(\ell)q(t) = 0$$

λ_n = wave length = $\frac{2\ell}{n}$

nodes = points at which $\hat{u}_n(x) = 0$



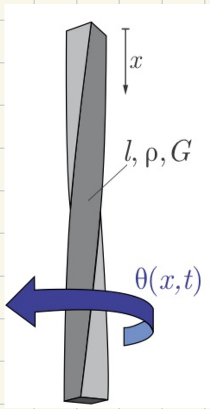
pinned-pinned eigenmodes

S.219

Torsional wave motion & vibrations

torsional wave equation for twisting of a homogeneous slender bar:

$$\ddot{\theta}(x,t) = \underbrace{c_T^2}_{\text{torsional wave speed}}\theta_{,xx}(x,t) \qquad \text{with} \qquad c_T = \sqrt{\frac{G}{\rho}}$$



S.220

Flexural wave motion & vibrations

flexural wave equation for bending of a homogeneous slender bar:

$$w_{,xxxx}(x,t) + \frac{\rho A}{EI_y}\ddot{w}(x,t) = 0$$

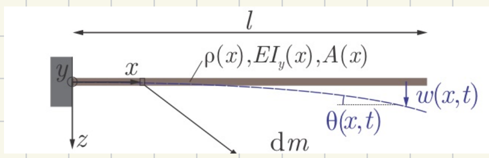
general solution for **flexural vibrations**:

$$w(x,t) = \hat{w}(x)q(t) \qquad \text{with} \qquad q(t) = A_1\cos(\omega t) + A_2\sin(\omega t)$$

and $\quad \hat{w}(x) = B_1\cos(kx) + B_2\sin(kx) + B_3\cosh(kx) + B_4\sinh(kx), \quad \omega^2 = k^4\frac{EI_y}{\rho A}$

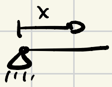
complete solution for **flexural vibrations**:

$$w(x,t) = \sum_{n=1}^{\infty}\hat{w}_n(x)q_n(t)$$

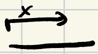


S.221

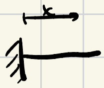
Generally:

1. 

$u(0) = 0 \quad u_x(0) \neq 0$

2. 

$u(0) \neq 0 \quad u_x(0) = 0$

3. 

$u(0) = 0 \quad u_x(0) = 0$